



OPEN ICT training with wearable devices to enhance emotional communication and musicality in vocal education

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The article examines the effectiveness of wearable technologies and mobile applications for developing musicality and vocal performance skills in vocal performance among first-year students of the Department of Vocal Singing at the Kharkiv National University of Arts named after I.P. Kotlyarevsky (N = 218). The study employed an eight-week training program incorporating the mobile application SingTrue alongside biometric data from wearable devices used to record physiological parameters to monitor breathing and heart rate during singing. The results of the study suggested that the training programme had improved the overall musicality of the first-year students from the experimental group ($p = 0.00$). It did not affect the ability to express emotions with voice ($p \geq 0.05$). Physiological data indicated changes in heart rate and respiration, reflecting an increase in physiological engagement during the singing process. The materials from this paper will be of interest to those who study the possibility of integrating mobile applications and complex software for musicians into areas such as higher music education. This paper draws attention to a limited number of scientific studies about the development of musicality of first-year students by using separate training programmes and applications, and emphasizes the need for further research aimed at the targeted development of emotional communication in vocal training.

Keywords Emotions, First year students, Interactive educational environment, Mobile learning, Music education, Singing, wearables, emotional communication, ICT training

In recent years, information technologies have been increasingly employed in the teaching of music and vocal performance¹. Tools such as multimedia, online resources, sound recording and analysis applications, among others, have become widely accessible to a broad audience². This accessibility suggests their potential to assist learners in developing musicality and even enhancing the emotional dimension of vocal performance^{3–5}. Previous research has primarily focused on the analysis of the acoustic properties of music and speech^{6,7}, yet has largely overlooked the significance of musicality and emotional expressiveness for professional musicians^{8,9}. Moreover, there is currently a notable shortage of studies addressing the integration of wearable devices—capable of capturing physiological and biomechanical indicators—into vocal training to support emotional communication^{10,11}. It is important to note that in the present study, emotional communication is conceptualized as a pedagogical construct referring to the ability to convey emotional meaning in vocal performance, whereas emotional expressiveness is used to denote measurable acoustic and physiological indicators. This article addresses the existing gap in research concerning the combined use of digital applications and wearable devices in vocal training, with particular emphasis on the development of musicality and emotional expressiveness. The scarcity of research in this area may be attributable to the inherent complexity of the subject, which poses significant challenges for scholars¹². These challenges include the difficulty of defining and measuring musicality and vocal emotions, as well as the need to account for cultural and individual differences in interpretation¹³. In addition, traditional methods of analyzing music and speech are constrained by limitations such as the inability to fully capture the dynamic nature of musical expression or to adequately incorporate the human factor^{14,15}.

At the same time, the assumption of a direct effect of digital technologies on the enhancement of emotional expressiveness in musical performance remains insufficiently substantiated. While robust evidence exists regarding their effectiveness in improving technical skills, auditory discrimination, and perceptual accuracy, empirical support for their impact on emotional expressiveness appears considerably less convincing^{1,3–5,16}. This is partly attributable to the nature of the relationship between music and emotion, which is characterized by

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conceptual complexity and the absence of a shared empirical consensus^{13,17,18}. Emotional content in music cannot be reduced solely to acoustic features or physiological responses, as it is shaped through interpretative decisions, stylistic frameworks, and culturally embedded communicative practices^{14,19,20}. Within this context, emotional expressiveness in vocal performance should be understood not as a coincidental by-product of technological influence, but as the outcome of intentional pedagogical design. From this perspective, the effectiveness of digital tools depends not only on their technical capabilities but also on the depth of their integration into an educational framework oriented toward the development of emotional communication as a core learning objective^{21–23}.

To address challenges, this article presents a study on how ICT-based training, enhanced by wearable devices providing real-time feedback derived from physiological and acoustic data, can be utilized to improve emotional communication and musicality in vocal education. It is anticipated that the combination of mobile applications with wearable technologies may enhance both the efficiency and personalization of the learning process^{24,25}. The practical implications of this research are considerable, as the ability to accurately convey emotions through music and speech is essential in various domains, including music education, the performing arts, and speech therapy.

The integration of wearable devices into ICT-based instruction opens new opportunities for evidence-based pedagogical strategies, enabling not only more precise measurement of emotional expressiveness but also the development of adaptive training programs tailored to learners' individual needs. This approach addresses a gap in the academic literature and establishes a foundation for future research in this field. The article emphasizes the importance of cultivating musicality and emotional vocal expression in singing while introducing an innovative instructional approach that combines information technologies with the enhanced capabilities of wearable devices.

Literature review

Information technologies play a pivotal role in the study of musicality and vocal emotions¹⁶. Researchers employ a variety of methods and technologies to analyze sounds, images, and performer behavior data in order to gain a more accurate understanding of their skills²⁶. These include the application of sound analysis techniques such as spectral analysis and formant analysis, as well as electromyography, to examine vocal emotions²⁷. In addition, studies have explored the use of computer vision to analyze body movements during musical performance^{28,29}. Examples are also provided of employing machine learning for the analysis of musicality and vocal emotions, describing the use of classification and regression algorithms to investigate the properties of musicality and emotional expressiveness in singing^{30,31}. Among these, deep learning algorithms and neural networks are utilized to categorize musical attributes such as tempo, timbre, melodic components, and harmonic structures, as well as to explore the emotional nuances of vocal expression^{32,33}. Furthermore, sound analyzers can process audio signals to determine various characteristics, including pitch, loudness, timbre, and duration³⁴. Modern applications also facilitate the study of musical characteristics and the emotional interpretation of music³⁵.

In recent years, wearable devices—such as smartwatches, respiratory monitors, and posture sensors—have been increasingly integrated into music and vocal research. These devices are used to record physiological parameters associated with emotional expression during singing¹¹. Such data provide an additional layer of information, enabling analysis of how emotional and physical states influence vocal performance and how singers convey emotions³⁶. Wearable technologies can also be valuable in developing adaptive learning strategies by delivering real-time feedback to both students and instructors³⁷. At the same time, it is essential to account for individual differences in musicality and vocal emotions⁸. Research has shown that specific brain regions—such as the posterior precentral gyrus, the posterior middle frontal gyrus, and the anterior inferior frontal gyrus—are activated during the performance of music with varying emotional qualities. Moreover, when listening to musical pieces that differ in tempo, the activity of certain brain areas—primarily the anterior central gyrus—changes accordingly^{17,38}. Thus, the perception of music and the emotional expression of singing are linked to the activity of distinct neural regions, a finding that can inform the development of innovative methods for training and educating singers and musicians¹⁸.

Mobile music applications can also serve as highly effective tools for developing the musical skills of students in higher music education institutions³⁹. Within this spectrum, examples include educational tools for music instruction, programs designed to enhance aural skills and sight-reading proficiency, platforms for music composition, and applications for recording musical fragments⁴⁰. Such resources can assist students in improving instrumental performance, refining musical hearing and theoretical knowledge, and composing original music⁴¹.

By creating synergy between wearable devices and these applications, it becomes possible to track physiological parameters during vocal lessons, which, in turn, may help instructors better monitor students' physiological responses and emotional engagement^{10,42}. At the same time, it is important to note certain theoretical divergences regarding which aspects of musicality and vocal emotions should be studied. Some researchers focus on the musical material and its characteristics, while others place greater emphasis on vocal technique and emotional interpretation^{43,44}.

Wearable devices introduce a new perspective into this debate by enabling data-driven analysis of the interplay between emotional expression and physical state, thereby contributing to a more holistic understanding of vocal training⁴⁵. Nonetheless, questions remain regarding the extent to which the obtained results may be subject to subjective interpretation—an issue relevant both to musical interpretation and to the evaluation of emotional expressiveness. Emotional expressiveness in singing is closely associated with the physiological and acoustic mechanisms of voice production and therefore cannot be examined in isolation. During vocal performance, the musical and emotional intention is conveyed by the singer through the regulation of pitch, phonatory characteristics, and dynamic variations throughout the vocal phrase. Research in vocal acoustics demonstrates

that expressive interpretation often involves subtle micro-adjustments of pitch and deviations from equal-tempered tuning, which may enhance the perceived emotional intensity of a performance⁴⁶.

In addition, acoustic properties of the singing voice, including spectral characteristics, vocal intensity, and phonatory patterns, influence listeners' perception of emotional expressiveness. This underscores that emotional communication in singing arises from the interaction of perceptual, physiological, and technical aspects of voice production⁴⁷.

At the same time, contemporary approaches in vocal pedagogy increasingly incorporate technological tools that provide real-time visual feedback on key acoustic and physiological parameters of the voice. Such technologies assist singers in developing greater awareness of breathing patterns, pitch stabilization, and vocal intensity levels, thereby facilitating a deeper understanding of the mechanisms underlying vocal performance. Real-time visual feedback is proposed as a valuable pedagogical resource that enhances awareness of the aerodynamic, physiological, and acoustic processes involved in voice production and supports the development of vocal control during training⁴⁸.

These considerations indicate that technological tools can not only complement traditional methods of vocal instruction but also facilitate the monitoring and improvement of the technical aspects that underpin expressive vocal performance.

Theoretical framework

In the present study, musicality is defined as a combination of auditory-perceptual and cognitive abilities that enable the perception and recognition of musical structures⁴⁹. Emotional communication in vocal performance is conceptualized as the capacity to convey the emotional content of musical material to the listener⁵⁰. Expressiveness is characterized as the degree to which a vocal performance is enriched with acoustic, intonational, and dynamic elements, reflecting the effectiveness of emotional transmission, yet remaining distinct from musicality¹⁸.

In the present study, physiological indicators such as heart rate and respiration are considered not as direct measures of emotional expressiveness but as indicators of physiological activity accompanying vocal performance. Within the context of automated vocal analysis, neutrality refers to a low manifestation of emotional features in the voice signal and functions as an indicator of limited vocal expression rather than as evidence of the absence of emotional experience⁵¹.

The current study is grounded in the integration of three interrelated domains: neurophysiological perspectives on emotions in musical performance, psycho-pedagogical approaches to emotional expressiveness and the communicative aspects of music, and the role of information and communication technologies in music education. From a neurophysiological standpoint, musical performance is accompanied by activation of the autonomic nervous system, manifested in changes in physiological indicators such as heart rate, respiration, and other parameters that signal the performer's emotional involvement⁵². These processes constitute a fundamental level of emotional response, providing a bodily basis for experiencing and regulating emotions during singing²⁰. At the same time, physiological changes alone do not ensure effective transmission of emotional content to the listener, as they require cognitive processing and interpretation⁵³.

The psycho-pedagogical perspective conceptualizes emotional expressiveness as the outcome of the purposeful development of performance skills, with particular emphasis placed on the social and cultural nature of emotions in the present context⁵⁴. Emotions are not only experienced by the performer but are also communicated through the vocal signal, thereby becoming accessible for perception by listeners¹⁴. Information and communication technologies extend this theoretical framework by demonstrating opportunities to enhance emotional communication through the recording and interpretation of physiological reactions and by supporting learning processes^{21,22}. By providing feedback on performance parameters, these tools stimulate the development of interpretative and communicative skills among learners⁵⁵. Consequently, technologies serve as a bridging element between the physiological dimensions of emotions and their conscious psycho-pedagogical interpretation²³.

Within the present study, wearable devices and digital applications are regarded not merely as instruments for registering physiological data but as components of a structured educational environment. Such integration is oriented toward fostering students' musical expressiveness and emotional communication skills and enables the alignment of data on neurophysiological activity with pedagogical objectives of vocal training. This approach supports a process that is meaningful both for the performer and for audience perception, linking empirical indicators of emotional engagement with the practical tasks of developing effective and perceptible vocal artistry.

Problem statement

The primary aim of this study is to empirically evaluate the effectiveness of an eight-week training program based on wearable technologies and information and communication tools (ICT), incorporating the SingTrue application and biometric indicators, for enhancing musicality and emotional communication in vocal performance among first-year students. To achieve this aim, three specific objectives were formulated:

- To assess changes in auditory perception, rhythmic accuracy, and the overall level of musicality in students following the intervention program, using the Advanced Measures of Music Audiation (AMMA) test.
- To compare the results of the experimental and control groups in order to determine the impact of the wearable technology and ICT-based training on musical skills.
- To analyze the dynamics of changes in students' ability to convey emotions in singing, based on voice analysis data (Vokaturi) and biometric indicators (heart rate and its variability, breathing rate).

Methods and materials

The SingTrue: Learn to Sing in Tune, Pitch Perfect Application is a mobile app for learning vocals and voice tuning on iOS. The application encompasses an array of interactive assignments designed to cultivate musicians' auditory acumen and ameliorate their vocal performance, encompassing vocal drills, timbre and sound differentiation exercises, as well as assessments of pitch modulation. It also provides feedback and performance statistics that singers can use to track their progress. Both novice singers and experienced musicians can use the application. Figure 1 presents the interface of the application.

In the application, pitch deviations were displayed relative to a target value through a simplified visual scale indicating pitch differences measured in cents (1/100 of a semitone). The visual indicator enabled participants to adjust their vocal output closer to the target pitch, and the primary purpose of this feedback was not to develop expressive pitch modulation but to improve intonation accuracy and stability during singing.

This paper used a tool such as the Advanced Measures of Music Audiation test^{57,58}. The test questions include both tonal and rhythmic measurements in the Advanced Measures of Music Audiation section. After listening to each of the 30 points (pairs), participants are asked to answer whether there is a difference in tone or rhythm between phrases. The test gives a “tonal assessment”, a “rhythmic assessment”, and a “general assessment”. Unlike other musicality tests, the AMMA one focuses on the ability to listen and understand musical structures such as rhythm, melody, harmony, and form. Each composition is evaluated according to three criteria: accuracy, speed, and confidence. The score is in points, and the maximum score is 40 points for each section. This means that each participant can score up to 80 points. The AMMA test was administered using standardized audio stimuli. Participants listened to pairs of short musical phrases presented through headphones in a noise-controlled classroom environment. After each pair, they were required to determine whether the phrases were similar or differed in either tonal or rhythmic structure. Thus, the test was designed to assess auditory perception and musical discrimination abilities rather than vocal performance skills.

No recordings of participants' vocal data were made during the testing procedure, as the AMMA test is specifically designed to evaluate musical abilities related to auditory perception.

Software VokatURI was used as a second metric tool. This is a software package intended to analyse the emotional and insight components of a singer's voice. At the same time, VokatURI, originally developed for speech analysis, was employed exclusively as a research-oriented analytical tool to identify acoustic patterns that may be associated with vocal expressiveness. This approach is based on the premise that low-level acoustic parameters—such as intensity dynamics, spectral balance, and pitch variability—can reflect characteristics of vocal expression. However, singing differs substantially from speech due to the extended duration of phonation, its distinctive melodic structure, and stylistic constraints. Therefore, the data obtained using VokatURI were interpreted as indicators of relative acoustic changes rather than as direct measurements of emotional communication. Previous studies have demonstrated that certain low-level acoustic parameters related to emotional expression, including intensity dynamics, spectral balance, and pitch variation, may also be relevant for vocal analysis. Nevertheless, singing differs fundamentally from speech due to the characteristics of sustained phonation, melodic structure, and stylistic constraints inherent to vocal performance^{59,60}. For this reason, within the present study, indicators derived from VokatURI data are interpreted with due caution and are considered exclusively in terms of relative

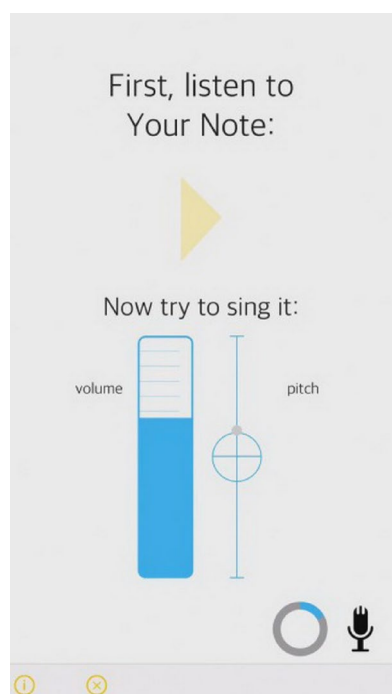


Fig. 1. SingTrue application⁵⁶. Source: App Store for iPhone

changes in the emotional expressiveness of the voice, rather than as markers of specific emotional states. The application uses machine vision training, which is based on neural networks. Figure 2 presents the application interface. In emotion detection, the “Neutrality” scale refers to the assessment of the level of expressiveness of singing, that is, the degree of its absence or presence. It covers all emotional states, including happiness, sadness, anger, and fear.

The study also employed wearable devices to provide supplementary biofeedback on the participants’ physical and emotional states. Smartwatches and chest sensors recorded breathing patterns, heart rate variability, and heart rate, with these data synchronized with the SingTrue and VokatURI applications to align vocal performance with physiological responses. The wearable devices delivered real-time feedback. Prior to each session, the devices were calibrated using the manufacturer’s standard initialization procedure to ensure the accuracy of the collected data. Physiological signals were recorded continuously throughout the performance of the vocal exercises. Data synchronization was conducted using recording timestamps: physiological parameters were continuously monitored during the vocal exercises, while audio recordings of the vocal excerpts were made in parallel. Because the recordings of physiological data and audio began and ended simultaneously, it was subsequently possible to analyze and align changes in physiological indicators with the corresponding segments of vocal performance. It is important to note that the physiological data in this study were used exclusively for analytical purposes, were not integrated into the educational program, and did not exert any direct influence on the music learning process.

Participants

This study involved the participation of 218 first-year students from the Department of Vocal Singing at Kharkiv National University of Arts, named after I.P. Kotlyarevsky. All of them were trained in vocal skills and were interested in improving their musicality and ability to express emotions through music. The control and experimental groups consisted of 109 people each. The average age of the participants was 18.54 years ($SD=0.11$). Prior to the intervention, all participants completed baseline assessments, including the AMMA test and physiological measurements recorded during vocal performance. The results were used to confirm the comparability of the control and experimental groups before their random allocation.

Study design

This training programme required the following equipment: a computer or laptop to download the application; a smartphone or tablet to install and use the application; musical instruments: a guitar, a piano, or other instruments necessary for singing in the right key; a metronome for maintaining rhythm and tempo; a tuner for tuning instruments and controlling singing tones; headphones for working with applications and tools necessary for sound effects and technologies; a sound processor for working with effects and sound technologies; wearable devices (smartwatches, chest sensors) for recording physiological parameters (heart rate and its variability, breathing patterns). Also, the authors of this article developed an 8-week training programme, which is presented in Fig. 3.

This sequence of exercises was developed by the authors of the present study based on existing modules within the SingTrue application. The program began with exercises aimed at developing fundamental auditory and intonational skills, including the ability to identify pitch and perceive rhythm.

Subsequent stages incorporated exercises focused on articulation, awareness of stylistic characteristics, and the expressive delivery of musical material. In the final weeks, instruction concentrated on the practical application of these skills, including singing with varying emotional coloration and preparation for stage performance.

The study design is characterized as quasi-experimental, as participants in the experimental group were able to use the application independently outside face-to-face classes without objective monitoring of actual usage time, while the control group did not receive a comparably intensive “placebo” intervention, because they

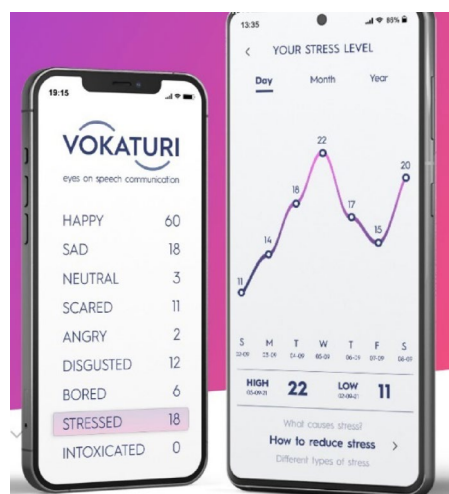


Fig. 2. VokatURIAndroid interface⁶¹. Source: VokatURI

Week	Training focus	Exercises from the app
1	Pitch detection and tonality testing	Pitch Perfector, Scale Trainer
2	Working on rhythm and tempo	Rhythm Master, Tempo Trainer
3	Improving articulation and diction	Articulation Master, Diction Trainer
4	Learning different styles and genres of music	Genre Explorer, Melody Matcher,
	Development of expressiveness and expression of	Harmony Hunter
5	emotions	Emotion Express, Vocal Range Extender,
6	Singing songs with different emotional shades and	Tone Tuner
	moods	Expressive Singing, Song Surgeon, Note
7	Preparing for a performance and working on interaction	Shifter
	with the audience	Performance Prep, Feedback Facilitator,
8	Assessing progress and planning further development	Crowd Pleaser
		Progress Tracker, Customized Challenges,
		Future Plans

Fig. 3. SingTrue training plan: development of musicality and emotional communication.

followed the standard educational program without additional technological intervention. These factors limit the rigor of causal inferences and reduce the strength of cause-and-effect conclusions. The participants were categorized into control and experimental groups using the random number technique within the Microsoft Excel software. The study involved first-year students from the Department of Vocal Performance who consented to participate and completed all stages of pre- and post-testing. Participation was voluntary, and all participants were enrolled in the same educational program and shared a comparable academic context, ensuring a consistent learning environment and equivalent instructional conditions. Students assigned to the control group followed the standard vocal training curriculum of the university’s vocal performance department during the same period. They attended their regular classes and practical training sessions organized by the university without participating in the additional educational program involving ICT, and without the use of the SingTrue application or wearable devices. Within this framework, the study aimed to evaluate differences between the traditional format of vocal instruction and the enhanced training model that incorporates digital tools and physiological monitoring.

The training program for students in the experimental group was organized to ensure comparable conditions for all participants, while allowing the instructor to adapt certain exercises to the individual characteristics of the students. For example, the instructor could adjust the pace of task completion, modify the number of repetitions, or increase the complexity of the musical material. In addition, the use of the SingTrue application allowed students to complete exercises at their own pace and repeat them as many times as necessary, which could partially mitigate differences in their levels of musical preparation. In the first stage, the participants of the experiment passed the Advanced Measures of Music Audiation test. The total test time was up to 60 min. Moreover, each student had to choose and record a song (up to 3 min long) to demonstrate their ability to express emotions through singing in the studio. They have to record the same song after the experiment. This means that it is impossible to change the song. To assess emotional expressiveness in singing, each participant recorded a self-selected composition in a studio setting. Identical microphones and recording equipment were used to ensure consistent acoustic conditions for all participants.

The resulting recordings were analyzed using the Vokaturi software library, which estimates the probabilities of different emotional states based on acoustic characteristics, including pitch variation, loudness levels, and spectral features. The results provided probabilistic scores for the primary emotional categories of joy, sadness, anger, fear, and neutral expression. Emotional expressiveness was evaluated through the distribution of these scores, with mean probability values calculated for each emotional category. This approach enabled comparisons of emotional expressiveness both across individual participants and between the control and experimental groups.

Participants’ vocal recordings were analyzed using Vokaturi at both the pre-test and post-test stages in order to obtain acoustic indicators reflecting vocal expressiveness. For each participant, two recordings were examined: one produced prior to the intervention and another recorded upon completion of the training program. After receiving the preliminary data, the experiment began. Only those first-year students who were randomly selected for the experimental group attended the training programme. Each lesson lasted 2 h, once a week, outside of school hours. Up to 10 people worked in real-time. During each session, participants wore sensors that recorded physiological parameters synchronized with vocal performance data, which were subsequently used to assess the relationship between physical state and emotional communication during singing. Smartwatches and chest-worn sensors were used exclusively during face-to-face classes within the experimental group, and their application followed a standardized protocol. The devices were fitted immediately before the performance of vocal exercises and recorded heart rate, heart rate variability, and respiration rate throughout the execution of the tasks. The collected physiological parameters were time-synchronized with vocal recordings and employed

as supplementary indicators of physiological involvement and emotional activation. At the same time, these data were not integrated into the instructional intervention and did not directly affect the principal measure of musicality (AMMA), which was assessed separately and independently. The participants had to perform the task autonomously on their smartphones.

However, they could receive information from a digital whiteboard, with the application broadcast on it during classes. There were no homework tasks. However, the participants could use the application not only during classes but also after them at their discretion. The actual extent of independent application use outside face-to-face classes, representing the intervention dosage, was not measured and was not included in the analysis. For the course to be more effective within the framework of classes, they could also use metronomes, musical instruments, and so on. The duration of the intervention (8 weeks) was selected as a realistic time frame consistent with the academic calendar and sufficient to encompass a complete cycle of weekly instruction, comprising eight sessions of two hours each. After the course, the participants from the control and experimental groups repeated the Advanced Measures of Music Audiation test. They recorded the songs they had chosen from the outset. Further, the researchers subjected the results to primary and secondary data processing.

Limitations

This study contained numerous limitations. The short duration of the developed programme is one of them. Moreover, the researchers offered the AMMA methodology to the respondents twice, with an interval of 2 months. This could affect the results. The students could regulate their activity by using the app after classes. Consequently, the level of extracurricular training and the use of the application could also affect the results, as well as the fact that different students had different initial competencies in the field of musicality and the ability to express feelings through voice. It should also be acknowledged that Vokaturi, originally developed for speech analysis and applied here to describe relative acoustic patterns of vocal expressiveness in singing, does not permit generalized conclusions about emotional communication in a broader psychological or aesthetic sense. Given that the effectiveness and reliability of speech-based emotion recognition systems when applied to singing remain insufficiently explored in the literature, the Vokaturi data in the present study should be interpreted primarily as illustrative indicators of acoustic patterns rather than as definitive measures of emotional expressiveness. The study was conducted within the framework of an existing educational program rather than under strictly controlled experimental conditions, which implies the influence of several contextual factors, such as individual performance style and prior interpretative experience. Moreover, the absence of a placebo condition or an alternative instructional method for the control group limits the ability to establish clear causal relationships. In addition, although participants were allowed to use the SingTrue application independently outside of scheduled sessions, the frequency and duration of such use were not systematically monitored. Consequently, the precise role of the application in shaping the observed outcomes remains uncertain.

Ethical issues

High ethics characterized this study. The experiment has previously passed the ethics commission at the Department of Vocal Singing, Kharkiv National University of Arts named after I.P. Kotlyarevsky. All the participants signed an informed consent form. In this way, they were aware of the actual goals and objectives of the study. There was no risk to the mental or physical health of the participants. They could terminate their participation in the experiment unilaterally at any time and stop attending classes. The companies mentioned in this paper were not interested parties and did not invest in this study. Kharkiv National University of Arts named after I.P. Kotlyarevsky bore all the expenses. The researchers did not benefit from the experiment and pursued exclusively research objectives. To address ethical considerations related to ensuring equal access to educational resources, students in the control group were granted access to SingTrue after the completion of the experimental phase of the study.

Data analysis

This paper used the Wilcoxon test for linked samples to determine whether there were statistically significant improvements before and after the experiment, as well as the non-parametric Mann–Whitney U-test for independent samples, which allowed the researchers to compare the control and experimental groups on such parameters as musicality and emotional expressiveness during singing. The data were processed utilizing the SPSS software package.

Results

Data analysis began with the processing of the results of the Advanced Measures of Music Audiation test. Table 1 shows the results of primary data processing.

The average value was 26.58 points on the first scale and 29.1 points on the second, respectively, if it is about the sample as a whole. The overall indicator is 55.60 points out of 80 points, combining the two previous scales. This relatively high result may be attributed to the fact that the study participants were novice musicians who possessed basic musical skills; however, these abilities still require further development through systematic training. Some intergroup differences are noticeable at the post-testing stage. The students who participated in the SingTrue: Development of Musicality and Emotional Communication programme showed, on average, higher success on the tonality scale. This indicator increased from 26.54 to 29.69 points. This was also true for the rhythm scale, where the average value increased from 28.95 points to 30.17 points. At the same time, there were no significant changes in the control group before and after the experiment.

Then the researchers compared the results of the groups to determine the reliability of the differences. Table 2 contains the results. The researchers found that there were no significant differences between the control and experimental groups at the pre-testing stage ($p \geq 0.05$).

	Experimental group			Control group			Total		
	Average value	N	Standard deviation	Average value	N	Standard deviation	Average value	N	Standard deviation
Tonal_pretest	26.54	109	2.275	26.62	109	2.280	26.58	218	2.273
Rhythm_pretest	28.95	109	1.443	29.07	109	1.392	29.01	218	1.416
Total_pretest	55.50	109	2.774	55.70	109	2.665	55.60	218	2.716
Tonal_posttest	29.69	109	1.714	27.16	109	2.157	28.42	218	2.321
Rhythm_posttest	30.17	109	1.407	29.09	109	1.405	29.63	218	1.504
Total_posttest	59.86	109	2.179	56.25	109	2.446	58.06	218	2.937

Table 1. Distribution of average values according to the AMMA method depending on the type of group.

	Tonal_pretest	Rhythm_pretest	Total_pretest	Tonal_posttest	Rhythm_posttest	Total_posttest
Mann-Whitney U	5,820.000	5,662.500	5,640.000	2,377.500	3,590.500	1,710.500
Wilcoxon's W	11,815.000	11,657.500	11,635.000	8,372.500	9,585.500	7,705.500
Z	−0.261	−0.609	−0.649	−7.718	−5.137	−9.138
Asymptotic significance (2-sided)	0.794	0.542	0.516	0.000	0.000	0.000

Table 2. The analysis of the statistical significance of intergroup differences by using the AMMA method.

	Tonal_posttest—Tonal_pretest	Rhythm_posttest—Rhythm_pretest	Total_posttest—Total_pretest
Z (the experimental group)	−7.730	−5.467	−8.219
Asymptotic significance (2-sided) (the experimental group)	0.000	0.000	0.000
Z (the control group)	−1.610	−0.071	−1.134
Asymptotic significance (2-sided) (the control group)	0.107	0.944	0.257

Table 3. The evaluation of the statistical significance of the differences before and after the experiment using the AMMA method.

This means that the random number method provided the conditions for conducting scientific research. At the same time, they found statistically significant differences between the groups on all three scales of the Advanced Measures of Music Audiation methodology ($p=0.00$). This gives reason to say that the 8-week training programme from this paper was effective in terms of developing the student's musicality.

In the next stage, the researchers carried out secondary data processing directly within the groups to confirm or refute the existing improvements. Table 3 contains the data for the group that attended the training course and for the control group.

Secondary data processing demonstrated that there are significant differences in the scales of tonality, rhythm, and overall score in the experimental group ($p=0.00$). This means that the developed programme allowed the students to strengthen their listening skills and improve their musicality. Also, there were no noticeable changes in the control group during the two months of the study. This proves that the developed training programme based on the SingTrue application was effective in improving the musicality of the first-year students who took part in the experiment. This result negates the fact that the first-year students passed the same test to assess their musical abilities over two months.

Next, the researchers processed the results, which were obtained after analysing the songs of the study participants in VokaturiAndroid. Figure 4 demonstrates the average values. The numbers were reduced to thousandths to simplify the perception of the text.

The examination of the primary data strongly indicated that the participating first-year students predominantly favoured the selection of songs characterized by a sense of joy. At the pre-test stage, the experimental group demonstrated a mean score of 0.50 on this scale. Participants were allowed to independently select a vocal piece for performance; consequently, some compositions appeared multiple times in the sample, as several students chose the same songs. At the same time, only 0.046 expressed sadness. Moreover, the researchers found that the students had not expressed emotions such as fear or anger through music (the average value was 0.00). This also resulted from the musical material that the researchers had chosen for the experiment. The musical material selected by the participants, which in many cases consisted of Chinese vocal repertoire, rarely required the expression of emotions such as anger or fear. The neutrality indicator averaged 0.489 in the experimental group. This means that the students still need to work a lot on expressiveness, which is average.

Joy also prevailed in the control group. The average value is 0.516. Sadness was conveyed much less often through music. The value was 0.041. Evaluating sensual expressiveness through singing, the neutrality indicator was 0.513 in this group at the preliminary testing stage.

The researchers recorded no significant changes in the mean values after the experiment. This indicates that the developed training programme was most likely ineffective in strengthening the skills of emotional expression

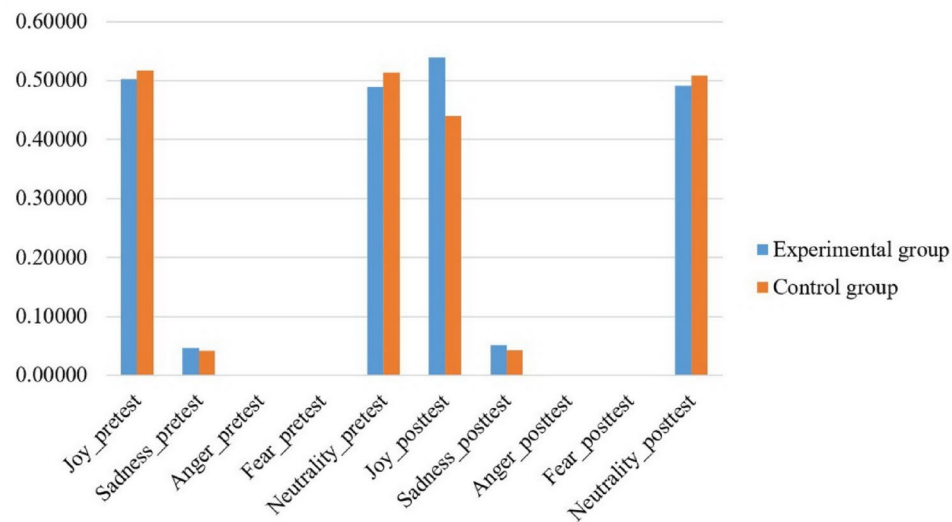


Fig. 4. Distribution of average values of VokatURIAndroid depending on the type of group.

	Mann–Whitney U	Wilcoxon’s W	Z	Asymptotic significance (2-sided)
Joy_pretest	5,739.000	11,734.000	– 0.433	0.665
Sadness_pretest	5266.000	11,261.000	– 1.449	0.147
Anger_pretest	5,940.500	11,935.500	0.000	1.000
Fear_pretest	5,940.500	11,935.500	0.000	1.000
Neutrality_pretest	5,684.000	11,679.000	– 0.551	0.582
Joy_posttest	4,703.000	10,698.000	– 2.658	0.008
Sadness_posttest	4,953.000	10,948.000	– 2.121	0.034
Anger_posttest	5,940.500	11,935.500	0.000	1.000
Fear_posttest	5,940.500	11,935.500	0.000	1.000
Neutrality_posttest	5,700.000	11,695.000	– 0.516	0.606
Heart Rate_pretest	5,720.000	11,420.000	– 0.482	0.701
Heart Rate Variability_pretest	4,520.000	10,120.000	– 2.081	0.042
Respiratory Rate_pretest	5,160.000	11,560.000	– 0.535	0.601
Heart Rate_posttest	4,780.000	10,380.000	– 2.232	0.032
Heart Rate Variability_posttest	5,810.000	11,410.000	– 0.120	0.911
Respiratory Rate_posttest	4,340.000	10,640.000	– 1.972	0.046

Table 4. The evaluation of the statistical significance of intergroup differences in the expression of emotional communication in vocal performance in VokatURIAndroid.

through music. This can be seen primarily on the neutrality scale, which is the main one in this paper. For example, the average value was 0.491 in the experimental group and 0.509 in the control group. Conversely, it is noteworthy for researchers to observe that the participants belonging to the control group exhibited a relatively diminished manifestation of elation on average in comparison to the preliminary testing phase (0.516 points versus 0.444). This may be due to some subjective factors, such as excitement or devaluation of participation in the study. It is necessary to consider the factor of a vocalist’s involvement when recording a song in a studio.

At the next stage, the researchers checked whether any changes had been significant before and after the experiment. Table 4 presents the results of secondary data processing.

As can be seen, the researchers found no significant differences between the control and experimental groups in terms of the intensity of emotion expression ($p \geq 0.05$). This means that the 8-week training program SingTrue: Development of Musicality and Emotional Communication programme failed to strengthen the skills of expressing emotions through the voice, which is associated with the ability to express emotions as intensely as possible in singing. At the same time, the analysis of wearable device data revealed that heart rate and breathing rate underwent statistically significant changes in the experimental group following the intervention ($p = 0.032$ and $p = 0.046$, respectively). In contrast, heart rate variability remained unchanged ($p = 0.911$), which may indicate the preservation of overall cardiovascular regulatory stability despite the increase in emotional expressiveness. These physiological indicators reflect increased physiological activation during singing; however, they should not be interpreted as direct indicators of emotional expressiveness.

Discussion

The results demonstrated improvement in musicality as measured by tonal, rhythmic, and overall AMMA scores, whereas no corresponding enhancement was observed in vocal emotional expressiveness assessed through automated acoustic analysis. The improvement in musicality is consistent with previous research reporting the effectiveness of mobile and ICT-based tools in developing auditory perception, intonation accuracy, and rhythmic sensitivity in music education^{4,5,41}. Within this conceptual framework, the observed gains in AMMA indicators align with the pedagogical orientation of the intervention and can be interpreted as a direct outcome of structured, technology-supported auditory training. An important aspect that should be considered concerns the orientation of the SingTrue application used in the study. The exercises included in the application are primarily designed to develop auditory perception, intonation accuracy, and pitch control rather than to enhance emotional expressiveness in singing. However, from an educational perspective, these technical components may be regarded as elements of vocal mastery that can indirectly influence expressive performance.

Improvements in pitch stability, rhythmic accuracy, and auditory awareness may enhance a singer's ability to shape musical phrases and convey interpretative nuances. Nevertheless, the results of the study suggest that the development of these perceptual and technical skills alone may be insufficient to produce substantial changes in the communication of emotional content through singing. This indicates that achieving emotional expressiveness requires specialized pedagogical approaches aimed at developing interpretative skills, emotional awareness, and the practice of expressive performance. In contrast, the absence of significant changes in vocal emotional expressiveness suggests that enhanced musicality does not automatically lead to stronger emotional communication in singing. This finding supports the theoretical distinction between perceptual and technical musical skills and expressive and interpretative capacities, which are shaped by different learning mechanisms and pedagogical approaches^{8,9}. Emotional communication in vocal performance involves not only control of acoustic parameters but also interpretative intention, stylistic awareness, and audience-oriented expressiveness, none of which were explicitly targeted within the present instructional program.

The lack of observable changes in acoustic indicators associated with emotions should also be interpreted in light of methodological limitations related to automated emotion recognition in singing. As discussed earlier, tools such as Vokaturi were originally developed for speech analysis and do not take into account musical structure, stylistic conventions, or performer intentions. Consequently, these indicators capture acoustic patterns linked to vocal expressiveness rather than the full complexity of emotional communication in vocal performance^{6,21,22}. Within these constraints, stability in emotion-related measures does not necessarily imply an absence of emotional involvement, but instead suggests that no clearly detectable changes occurred at the specific acoustic level targeted by the software.

Physiological data collected using wearable devices revealed statistically significant changes in heart rate and respiration in the experimental group following the intervention, while heart rate variability remained stable. However, these indicators should not be interpreted as direct measures of the emotional expressiveness of vocal performance, as fluctuations in heart rate and respiration during singing may be explained by multiple factors, including physical exertion, breath control, concentration, performance-related anxiety, fatigue, or the complexity of the musical material^{10,11}. Consequently, the observed changes are more likely associated with an increased level of physiological activation during vocal activity rather than with the direct manifestation of emotional communication. The stability of heart rate variability indicates that the overall balance of autonomic regulation was maintained, which further supports the interpretation that physiological activation was not linked to qualitative changes in emotional regulation or expression. These conclusions are consistent with findings from previous studies suggesting that physiological responses to music and performance may occur independently of overt emotional expression^{15,18}. Although one study has reported that biofeedback and emotion-focused interventions may influence emotional regulation^{62–64}, such effects require clearly articulated training components aimed at emotional awareness and control, which were not included in the present intervention program.

Comparison with studies employing machine learning and deep learning approaches to emotion recognition further highlights the specificity of the obtained results. Research reporting high classification accuracy based on acoustic and multimodal features^{31–33} typically relies on controlled datasets, explicit emotion labels, or indicators derived from song lyrics. None of these components was incorporated into the present study. The findings emphasize the importance of aligning technological tools with clearly defined pedagogical objectives and point to the need for future investigations aimed at addressing the limitations of the current research.

Conclusion

The results indicate that the eight-week instructional program incorporating ICT and the SingTrue application successfully enhanced the musicality of first-year vocal students, as evidenced by growth in tonal, rhythmic, and overall AMMA scores in the experimental group compared to the control group. The findings indicate that targeted, technology-supported instruction can facilitate the development of perceptual and auditory components of musical competence within a relatively short period of time. At the same time, the intervention did not yield a statistically significant effect on emotional communication in vocal performance, suggesting that improvements in musical perception and technical accuracy do not necessarily translate into enhanced emotional conveyance in singing. Physiological data obtained via wearable devices recorded significant changes in heart rate and respiration rate in the experimental group following the intervention, while heart rate variability remained unchanged, pointing to increased physiological activation and involvement in the process of vocal performance. However, these measures cannot be regarded as direct evidence of enhanced emotional expressiveness. The primary contribution of the study lies in clarifying the conditions under which technologies can be effective in vocal training: in the present case, they support the development of musicality, whereas

their impact on emotional communication remains limited in the absence of explicit integration of expressive dimensions into the overall pedagogical approach. This underscores the need to move beyond standardized technological solutions toward the design of instructional programs that place deliberate emphasis on the cultivation of emotional expressiveness. It is important that future research focus on longitudinal interventions, the development of emotion assessment methods tailored to musical performance, and pedagogical approaches that emphasize emotional expression and communication in singing. Further investigations should also address motivational factors related to emotional expressiveness in artistic contexts and explore how integrating these elements into digital applications may facilitate the advancement of emotional expressiveness in music and vocal education.

Data availability

All data generated or analysed during this study are included in this published article.

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Author contributions

Zhang Ling: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing—original draft, Writing—review & editing.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

The experiment has previously passed the ethics commission at the Department of Vocal Singing, Kharkiv National University of Arts named after I.P. Kotlyarevsky (Protocol No. 681 of 02/02/2024). The study was conducted in accordance with the rules of the Declaration of Helsinki.

Informed consent

All the participants signed an informed consent form.

Additional information

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